Light

GCSE Computer Science – Searching and Sorting Algorithms (Set 3)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

How does linear search find a target value in a list?

- ○ **A.** By jumping to random positions until the target is found
- ○ **B.** By starting at the middle and eliminating half the list each time
- ○ **C.** By sorting the list first, then searching
- ○ **D.** By checking each item in sequence from the start until the target is found or the list ends

Q2 (1 mark)

What must be true about a list before binary search can be used?

- ○ **A.** The list must be sorted in order
- ○ **B.** The list must contain only numbers
- ○ **C.** The list must have fewer than 100 items
- ○ **D.** The list must contain no duplicate values

Q3 (1 mark)

What operation does bubble sort repeatedly perform?

- ○ **A.** Dividing the list into two halves
- ○ **B.** Comparing adjacent pairs and swapping them if they are in the wrong order
- ○ **C.** Finding the smallest item and moving it to the start
- ○ **D.** Randomly shuffling items until sorted

Q4 (1 mark)

What approach does merge sort use to sort a list?

- ○ **A.** Repeatedly swapping adjacent items
- ○ **B.** Finding the maximum value first
- ○ **C.** Divide and conquer - splitting the list into smaller parts, sorting them, then merging back together
- ○ **D.** Checking each item against every other item

Q5 (1 mark)

What is one advantage of linear search over binary search?

- ○ **A.** Linear search works on unsorted lists
- ○ **B.** Linear search is always faster
- ○ **C.** Linear search uses less memory than binary search
- ○ **D.** Linear search can only find one item

Q6 (1 mark)

Why is binary search efficient on large sorted lists?

- ○ **A.** Because it eliminates half of the remaining items with each comparison
- ○ **B.** Because it checks every item twice
- ○ **C.** Because it uses more memory
- ○ **D.** Because it only works on small lists

Q7 (1 mark)

What is guaranteed after one complete pass of bubble sort on an ascending sort?

- ○ **A.** The entire list is sorted
- ○ **B.** The smallest item is in the first position
- ○ **C.** The largest item is in the last position

- ○ **D.** Half the items are sorted

Q8 (1 mark)

In merge sort, why does the splitting stop when sublists contain only one item?

- ○ **A.** Because the computer runs out of memory
- ○ **B.** Because single items cannot be divided further
- ○ **C.** Because a list with one item is already sorted - there is nothing to put in wrong order
- ○ **D.** Because single items are deleted during sorting

Q9 (1 mark)

When comparing algorithm efficiency at GCSE level, what factor is primarily considered?

- ○ **A.** Time efficiency - how long the algorithm takes to complete
- ○ **B.** How much electricity the algorithm uses
- ○ **C.** How colourful the output appears
- ○ **D.** The programming language used

Q10 (1 mark)

In the best case scenario, how many comparisons does linear search need to find an item?

- ○ **A.** Zero comparisons
- ○ **B.** Half the list length
- ○ **C.** The entire list length
- ○ **D.** One comparison

Q11 (1 mark)

What is one advantage of bubble sort?

- ○ **A.** It is always the fastest sorting algorithm
- ○ **B.** It is simple to understand and implement
- ○ **C.** It uses the least memory of any sorting algorithm
- ○ **D.** It can only sort numbers, not text

Q12 (1 mark)

What is one disadvantage of merge sort compared to bubble sort?

- ○ **A.** Merge sort is always slower than bubble sort
- ○ **B.** Merge sort requires additional memory to store temporary sublists during merging
- ○ **C.** Merge sort cannot sort text data
- ○ **D.** Merge sort only works on lists with even numbers of items

Q13 (2 marks)

A linear search is performed on the list [12, 7, 25, 3, 18] to find the value 25. How many comparisons are made?

- ○ **A.** 1 comparison
- ○ **B.** 2 comparisons
- ○ **C.** 5 comparisons
- ○ **D.** 3 comparisons

Q14 (2 marks)

A binary search is performed on the sorted list [5, 12, 19, 26, 33, 40, 47] to find 40. After checking the middle item, which half does the search continue in?

- ○ **A.** The left half [5, 12, 19]
- ○ **B.** The right half [33, 40, 47]
- ○ **C.** Both halves simultaneously
- ○ **D.** The search restarts from the beginning

Q15 (2 marks)

After one complete pass of bubble sort (ascending) on the list [6, 2, 8, 1], what is the state of the list?

- ○ **A.** [1, 2, 6, 8]
- ○ **B.** [2, 6, 1, 8]

- **C.** [2, 1, 6, 8]
- **D.** [6, 2, 1, 8]

Q16 (2 marks)

When merge sort divides the list [9, 3, 7, 1, 5, 8] into single items, how many levels of splitting occur?

- **A.** 2 levels
- **B.** 6 levels
- **C.** 4 levels
- **D.** 3 levels

Q17 (2 marks)

A company has 8,000 customer records stored in alphabetical order. Why would binary search be more efficient than linear search for finding a specific customer?

- **A.** Binary search would check at most about 13 records, while linear search might check all 8,000
- **B.** Binary search alphabetises the records automatically
- **C.** Linear search cannot work with customer records
- **D.** Both would check the same number of records

Q18 (2 marks)

How can bubble sort detect that a list is already sorted?

- ○ **A.** By counting the number of items in the list
- ○ **B.** By checking if the first item equals the last item
- ○ **C.** If a complete pass is made with no swaps, the list must be sorted
- ○ **D.** It cannot detect this - it always does the same number of passes

Q19 (2 marks)

When merging the sorted sublists [4, 11] and [6, 9] in merge sort, what is the first item placed in the merged result?

- ○ **A.** 11 (the largest item)
- ○ **B.** 4 (the smallest item from the first sublist compared with the first item of the second)
- ○ **C.** 6 (the first item of the second sublist)
- ○ **D.** 9 (the last item alphabetically)

Q20 (2 marks)

A music app maintains a playlist where users frequently add songs in random order. Which search algorithm is most practical for finding a specific song?

- ○ **A.** Binary search because it is always faster
- ○ **B.** Linear search because the playlist is not sorted and sorting would be impractical
- ○ **C.** Neither algorithm can search a playlist

- ○ **D.** Both algorithms would be equally efficient

Q21 (2 marks)

What is the maximum number of passes bubble sort might need to fully sort a list of 8 items?

- ○ **A.** 8 passes
- ○ **B.** 7 passes
- ○ **C.** 16 passes
- ○ **D.** 64 passes

Q22 (2 marks)

In binary search, when searching a list with items at indices 0 to 10, which index is checked first?

- ○ **A.** Index 0
- ○ **B.** Index 10
- ○ **C.** Index 5
- ○ **D.** Index 6

Q23 (2 marks)

Why is merge sort generally faster than bubble sort for large datasets?

- ○ **A.** Merge sort uses simpler comparisons

- ○ **B.** Merge sort's divide-and-conquer approach makes fewer total comparisons as data size grows
- ○ **C.** Bubble sort cannot handle large datasets at all
- ○ **D.** Merge sort skips comparisons randomly

Q24 (3 marks)

Perform a complete binary search on the sorted list [3, 8, 15, 22, 29, 36, 43, 50] to find 36. How many comparisons are needed?

- ○ **A.** 1 comparison
- ○ **B.** 8 comparisons
- ○ **C.** 2 comparisons
- ○ **D.** 3 comparisons

Q25 (3 marks)

How many total swaps are needed to completely sort the list [5, 1, 4, 2] using bubble sort (ascending)?

- ○ **A.** 3 swaps
- ○ **B.** 4 swaps
- ○ **C.** 5 swaps
- ○ **D.** 6 swaps

Q26 (3 marks)

During merge sort on the list [7, 2, 5, 3], what are the two sorted sublists immediately before the final merge?

- ○ **A.** [7, 2] and [5, 3]
- ○ **B.** [2, 3] and [5, 7]
- ○ **C.** [7] and [2, 3, 5]
- ○ **D.** [2, 7] and [3, 5]

Q27 (3 marks)

For a sorted list of 128 items, compare the worst-case comparisons for linear search versus binary search.

- ○ **A.** Linear: 128, Binary: 7
- ○ **B.** Linear: 64, Binary: 64
- ○ **C.** Linear: 128, Binary: 128
- ○ **D.** Linear: 7, Binary: 128

Q28 (3 marks)

A hospital needs to sort 500 patient records by surname for an urgent report. The records are currently in random order. Analyse which sorting algorithm would be more suitable.

- ○ **A.** Bubble sort, because it is simpler to understand
- ○ **B.** Neither algorithm can sort text records
- ○ **C.** Both algorithms would take exactly the same time

○ **D.** Merge sort, because it requires far fewer comparisons for 500 randomly-ordered items

Q29 (3 marks)

A programmer uses binary search on the unsorted list [15, 8, 22, 4, 19] to find the value 8. The algorithm reports '8 not found'. Explain why this incorrect result occurs.

○ **A.** Binary search cannot search for single-digit numbers

○ **B.** The list is too short for binary search to work

○ **C.** Binary search requires sorted data; on unsorted data, it may search the wrong half and miss items that are actually present

○ **D.** The number 8 is at an even index, which binary search skips

Q30 (3 marks)

A school maintains an exam results database with 2,000 students sorted alphabetically by surname. Results rarely change, but teachers frequently search for specific students. Which search algorithm should be used, and why?

○ **A.** Linear search, because it is simpler to implement

○ **B.** Binary search, because the data is sorted, searches are frequent, and binary search needs only about 11 comparisons versus linear search's potential 2,000

○ **C.** Neither algorithm is suitable for educational data

○ **D.** Random search, because student names are unpredictable

Total: **55** marks • Time: **30 min**

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